

Basel II

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Introduction

1. The fundamental objective of the Committee's work to revise the 1988 Accord² has been to develop a framework that would further strengthen the soundness and stability of the international banking system while maintaining sufficient consistency that capital adequacy regulation will not be a significant source of competitive inequality among internationally active banks. The Committee notes that, in their comments on the proposals, banks and other interested parties have welcomed the concept and rationale of the three pillars (minimum capital requirements, supervisory review, and market discipline) approach on which the revised Framework is based.
2. In developing the revised Framework, the Committee has sought to arrive at significantly more risk-sensitive capital requirements that are conceptually sound and at the same time pay due regard to particular features of the present supervisory and accounting systems in individual member countries. It believes that this objective has been achieved. The Committee is also retaining key elements of the 1988 capital adequacy framework, including the general requirement for banks to hold total capital equivalent to at least 8% of their risk-weighted assets; the basic structure of the 1996 Market Risk Amendment regarding the treatment of market risk; and the definition of eligible capital.
3. The revised Framework provides a range of options for determining the capital requirements for credit risk and operational risk to allow banks and supervisors to select approaches that are most appropriate for their operations and their financial market infrastructure.
4. It should be stressed that the revised Framework is designed to establish minimum levels of capital for internationally active banks. As under the 1988 Accord, national authorities will be free to adopt arrangements that set higher levels of minimum capital. Moreover, they are free to put in place supplementary measures of capital adequacy for the banking organisations they charter.
5. The Committee is aware that interactions between regulatory and accounting approaches at both the national and international level can have significant consequences for the comparability of the resulting measures of capital adequacy and for the costs associated with the implementation of these approaches. The Committee believes that its decisions with respect to unexpected and expected losses represent a major step forward in this regard. The Committee and its members intend to continue playing a pro-active role in the dialogue with accounting authorities in an effort to reduce, wherever possible, inappropriate disparities between regulatory and accounting standards.

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Part 1: Scope of Application

V. Significant investments in commercial entities

1. Significant minority and majority investments in commercial entities which exceed certain materiality levels will be deducted from banks' capital. Materiality levels will be determined by national accounting and/or regulatory practices. Materiality levels of 15% of the bank's capital for individual significant investments in commercial entities and 60% of the bank's capital for the aggregate of such investments, or stricter levels, will be applied. The amount to be deducted will be that portion of the investment that exceeds the materiality level.
2. Investments in significant minority- and majority-owned and -controlled commercial entities below the materiality levels noted above will be risk-weighted at no lower than 100% for banks using the standardised approach. For banks using the IRB approach, the investment would be risk weighted in accordance with the methodology the Committee is developing for equities and would not be less than 100%.

VI. Deduction of investments pursuant to this part

1. Where deductions of investments are made pursuant to this part on scope of application, the deductions will be 50% from Tier 1 and 50% from Tier 2 capital.
2. Goodwill relating to entities subject to a deduction approach pursuant to this part should be deducted from Tier 1 in the same manner as goodwill relating to consolidated subsidiaries, and the remainder of the investments should be deducted as provided for in this part. A similar treatment of goodwill should be applied, if using an alternative group-wide approach pursuant to paragraph 30.
3. The limits on Tier 2 and Tier 3 capital and on innovative Tier 1 instruments will be based on the amount of Tier 1 capital after deduction of goodwill but before the deductions of investments pursuant to this part on scope of application (see Annex 1 for an example how to calculate the 15% limit for innovative Tier 1 instruments).

Part 2: The First Pillar – Minimum Capital Requirements

I. Calculation of minimum capital requirements

1. Part 2 presents the calculation of the total minimum capital requirements for credit, market and operational risk. The capital ratio is calculated using the definition of regulatory capital and risk-weighted assets. The total capital ratio must be no lower than 8%. Tier 2 capital is limited to 100% of Tier 1 capital.

A. Regulatory capital

1. The definition of eligible regulatory capital, as outlined in the 1988 Accord and clarified in the 27 October 1998 press release on "Instruments eligible for inclusion in Tier 1 capital", remains in place except for the modifications in paragraphs 37 to 39 and 43. The definition is outlined in paragraphs 49 (i) to 49 (xviii) and in Annex Ia.
2. Under the standardised approach to credit risk, general provisions, as explained in paragraphs 381 to 383, can be included in Tier 2 capital subject to the limit of 1.25% of riskweighted assets.
3. Under the internal ratings-based (IRB) approach, the treatment of the 1988 Accord to include general provisions (or general loan-loss reserves) in Tier 2 capital is withdrawn. Banks using the IRB approach for securitisation exposures or the PD/LGD approach for equity exposures must first deduct the EL amounts subject to the corresponding conditions in paragraphs 563 and 386, respectively. Banks using the IRB approach for other asset classes must compare (i) the amount of total eligible provisions, as defined in paragraph 380, with (ii) the total expected losses (EL) amount as calculated within the IRB approach and defined in paragraph 375. Where the total expected loss (EL) amount exceeds total eligible provisions, banks must deduct the difference. Deduction must be on the basis of 50% from Tier 1 and 50% from Tier 2. Where the total expected loss (EL) amount is less than total eligible provisions, as explained in paragraphs 380 to 383, banks may recognise the difference in Tier 2 capital up to a maximum of 0.6% of credit risk-weighted assets. At national discretion, a limit lower than 0.6% may be applied.

B. Risk-weighted assets

1. Total risk-weighted assets are determined by multiplying the capital requirements for market risk and operational risk by 12.5 (i.e. the reciprocal of the minimum capital ratio of 8%) and adding the resulting figures to the sum of risk-weighted assets for credit risk. The Committee applies a scaling factor in order to broadly maintain the aggregate level of minimum capital requirements, while also providing incentives to adopt the more advanced risk-sensitive approaches of the Framework. The current best estimate of the scaling factor is 1.06. National authorities will continue to monitor capital requirements during the implementation period of this Framework. Moreover, the Committee will monitor national experiences with this Framework. The scaling factor is applied to the riskweighted asset amounts for credit risk assessed under the IRB approach.

Ia. The constituents of capital

A. Core capital (basic equity or Tier 1)

49 (i). The Committee considers that the key element of capital on which the main emphasis should be placed is equity capital (issued and fully paid ordinary shares/common stock and non-cumulative perpetual preferred stock but excluding cumulative preferred stock) and disclosed reserves. This key element of capital is the only element common to all countries' banking systems; it is wholly visible in the published accounts and is the basis on which most market judgements of capital adequacy are made; and it has a crucial bearing on profit margins and a bank's ability to compete. This emphasis on equity capital and disclosed reserves reflects the importance the Committee attaches to securing an appropriate quality, and the level, of the total capital resources maintained by major banks.

49 (iii). The Committee has therefore concluded that capital, for supervisory purposes, should be defined in two tiers in a way which will have the effect of requiring at least 50% of a bank's capital base to consist of a core element comprised of equity capital and published reserves from post-tax retained earnings (Tier 1). The other elements of capital (supplementary capital) will be admitted into Tier 2 limited to 100% of Tier 1. These supplementary capital elements and the particular conditions attaching to their inclusion in the capital base are set out in paragraphs 49(iv) to 49(xii) below and in more detail in Annex 1a. Each of these elements may be included or not included by national authorities at their discretion in the light of their national accounting and supervisory regulations.

B. Supplementary capital (Tier 2)

1. Undisclosed reserves
2. Revaluation reserves
3. General provisions/general loan-loss reserves

49(vii). General provisions or general loan-loss reserves are created against the possibility of losses not yet identified. Where they do not reflect a known deterioration in the valuation of particular assets, these reserves qualify for inclusion in Tier 2 capital. Where, however, provisions or reserves have been created against identified losses or in respect of an identified deterioration in the value of any asset or group of subsets of assets, they are not freely available to meet unidentified losses which may subsequently arise elsewhere in the portfolio and do not possess an essential characteristic of capital. Such provisions or reserves should therefore not be included in the capital base.

1. Hybrid debt capital instruments
2. Subordinated term debt

49(xii). The Committee is agreed that subordinated term debt instruments have significant deficiencies as constituents of capital in view of their fixed maturity and inability to absorb losses except in a liquidation. These deficiencies justify an additional restriction on the amount of such debt capital which is eligible for inclusion within the capital base. Consequently, it has been concluded that subordinated term debt instruments with a minimum original term to maturity of over five years may be included within the supplementary elements of capital, but only to a maximum of 50% of the core capital element and subject to adequate amortisation arrangements.

C. Short-term subordinated debt covering market risk (Tier 3)

49(xiii). The principal form of eligible capital to cover market risks consists of shareholders' equity and retained earnings (Tier 1 capital) and supplementary capital (Tier 2 capital) as defined in paragraphs 49(i) to 49(xii). But banks may also, at the discretion of their national authority, employ a third tier of capital ("Tier 3"), consisting of short-term subordinated debt as defined in paragraph 49(xiv) below for the sole purpose of meeting a proportion of the capital requirements for market risks, subject to the following conditions:

- Banks will be entitled to use Tier 3 capital solely to support market risks as defined in paragraphs 709 to 718(Lxix). This means that any capital requirement arising in respect of credit and counterparty risk in the terms of this Framework, including the credit counterparty risk in respect of OTCs and SFTs in both trading and banking books, needs to be met by the existing definition of capital base set out in paragraphs 49(i) to 49(xii) above (i.e. Tiers 1 and 2);

- Tier 3 capital will be limited to 250% of a bank's Tier 1 capital that is required to support market risks. This means that a minimum of about 28½% of market risks needs to be supported by Tier 1 capital that is not required to support risks in the remainder of the book;
- Tier 2 elements may be substituted for Tier 3 up to the same limit of 250% in so far as the overall limits set out in paragraph 49(iii) above are not breached, that is to say eligible Tier 2 capital may not exceed total Tier 1 capital, and long-term subordinated debt may not exceed 50% of Tier 1 capital;

49(xiv). For short-term subordinated debt to be eligible as Tier 3 capital, it needs, if circumstances demand, to be capable of becoming part of a bank's permanent capital and thus be available to absorb losses in the event of insolvency. It must, therefore, at a minimum:

- be unsecured, subordinated and fully paid up;
- have an original maturity of at least two years;
- not be repayable before the agreed repayment date unless the supervisory authority agrees;
- be subject to a lock-in clause which stipulates that neither interest nor principal may be paid (even at maturity) if such payment means that the bank falls below or remains below its minimum capital requirement.

D. Deductions from capital

49(xv). It has been concluded that the following deductions should be made from the capital base for the purpose of calculating the risk-weighted capital ratio. The deductions will consist of:

- (i) Goodwill, as a deduction from Tier 1 capital elements;
- (ii) Increase in equity capital resulting from a securitisation exposure, as a deduction from Tier 1 capital elements, pursuant to paragraph 562 below;
- Investments in subsidiaries engaged in banking and financial activities which are not consolidated in national systems.

II. Credit Risk – The Standardised Approach

50. The Committee permits banks a choice between two broad methodologies for calculating their capital requirements for credit risk. One alternative, the Standardised Approach, will be to measure credit risk in a standardised manner, supported by external credit assessments.

51. The other alternative, the Internal Ratings-based Approach, which is subject to the explicit approval of the bank's supervisor, would allow banks to use their internal rating systems for credit risk.

52. The following section sets out revisions to the 1988 Accord for risk weighting banking book exposures. Exposures that are not explicitly addressed in this section will retain the current treatment; however, exposures related to securitisation are dealt with in Section IV. Exposures should be risk-weighted net of specific provisions.

A. Individual claims

1. Claims on sovereigns

53. Claims on sovereigns and their central banks will be risk weighted as follows:

Credit Assessment	AAA to AA-	A+ to A-	BBB+ to BBB-	BB+ to B-	Below B-	Unrated									
Unrated Risk Weight	0%	20%	50%	100%	150%	100%									

54. At national discretion, a lower risk weight may be applied to banks' exposures to their sovereign (or central bank) of incorporation denominated in domestic currency and funded¹⁹ in that currency.

2. Claims on non-central government public sector entities (PSEs)

56. Claims on the Bank for International Settlements, the International Monetary Fund, the European Central Bank and the European Community may receive a 0% risk weight.
57. Claims on domestic PSEs will be risk-weighted at national discretion, according to either option 1 or option 2 for claims on banks.
58. Subject to national discretion, claims on certain domestic PSEs may also be treated as claims on the sovereigns in whose jurisdictions the PSEs are established.

3. Claims on multilateral development banks (MDBs)

59. The risk weights applied to claims on MDBs will generally be based on external credit assessments as set out under option 2 for claims on banks but without the possibility of using the preferential treatment for short-term claims. A 0% risk weight will be applied to claims on highly rated MDBs that fulfil to the Committee's satisfaction the criteria provided below.

4. Claims on banks

60. There are two options for claims on banks. National supervisors will apply one option to all banks in their jurisdiction. No claim on an unrated bank may receive a risk weight lower than that applied to claims on its sovereign of incorporation.
61. Under the first option, all banks incorporated in a given country will be assigned a risk weight one category less favourable than that assigned to claims on the sovereign of that country. However, for claims on banks in countries with sovereigns rated BB+ to B- and on banks in unrated countries the risk weight will be capped at 100%.
62. The second option bases the risk weighting on the external credit assessment of the bank itself with claims on unrated banks being risk-weighted at 50%. Under this option, a preferential risk weight that is one category more favourable may be applied to claims with an original maturity of three months or less, subject to a floor of 20%. This treatment will be available to both rated and unrated banks, but not to banks risk weighted at 150%.
63. The two options are summarised in the tables below.

Option 1

Credit Assessment of Sovereign	AAA to AA-	A+ to A-	BBB+ to BBB-	BB+ to B-	Below B-	Unrated									
Option 1 Risk Weight	20%	50%	100%	100%	150%	100%									

Option 2

Credit Assessment of Banks	AAA to AA-	A+ to A-	BBB+ to BBB-	BB+ to B-	Below B-	Unrated									
Option 2 Risk Weight	20%	50%	50%	100%	150%	50%									
ST claims Risk Weight	20%	20%	20%	50%	150%	20%									

6. Claims on corporates

66. The table provided below illustrates the risk weighting of rated corporate claims, including claims on insurance companies. The standard risk weight for unrated claims on corporates will be 100%. No claim on an unrated corporate may be given a risk weight preferential to that assigned to its sovereign of incorporation.

Credit Assessment of Corporate	AAA to AA-	A+ to A-	BBB+ to BB-	Below BB-	Unrated
Risk Weight	20%	50%	100%	150%	100%

67. Supervisory authorities should increase the standard risk weight for unrated claims where they judge that a higher risk weight is warranted by the overall default experience in their jurisdiction. As part of the supervisory review process, supervisors may also consider whether the credit quality of corporate claims held by individual banks should warrant a standard risk weight higher than 100%.

68. At national discretion, supervisory authorities may permit banks to risk weight all corporate claims at 100% without regard to external ratings. Where this discretion is exercised by the supervisor, it must ensure that banks apply a single consistent approach, i.e. either to use ratings wherever available or not at all. To prevent “cherry-picking” of external ratings, banks should obtain supervisory approval before utilising this option to risk weight all corporate claims at 100%.

7. Claims included in the regulatory retail portfolios

69. Claims that qualify under the criteria listed in paragraph 70 may be considered as retail claims for regulatory capital purposes and included in a regulatory retail portfolio. Exposures included in such a portfolio may be risk-weighted at 75%, except as provided in paragraph 75 for past due loans.

70. To be included in the regulatory retail portfolio, claims must meet the following four criteria:

- Orientation criterion — The exposure is to an individual person or persons or to a small business;
- Product criterion — The exposure takes the form of any of the following: revolving credits and lines of credit (including credit cards and overdrafts), personal term loans and leases (e.g. instalment loans, auto loans and leases, student and educational loans, personal finance) and small business facilities and commitments. Securities (such as bonds and equities), whether listed or not, are specifically excluded from this category. Mortgage loans are excluded to the extent that they qualify for treatment as claims secured by residential property (see paragraph 72).
- Granularity criterion — The supervisor must be satisfied that the regulatory retail portfolio is sufficiently diversified to a degree that reduces the risks in the portfolio, warranting the 75% risk weight. One way of achieving this may be to set a numerical limit that no aggregate exposure to one counterpart²⁸ can exceed 0.2% of the overall regulatory retail portfolio.
- Low value of individual exposures. The maximum aggregated retail exposure to one counterpart cannot exceed an absolute threshold of €1 million.

71. National supervisory authorities should evaluate whether the risk weights in paragraph 69 are considered to be too low based on the default experience for these types of exposures in their jurisdictions. Supervisors, therefore, may require banks to increase these risk weights as appropriate.

8. Claims secured by residential property

72. Lending fully secured by mortgages on residential property that is or will be occupied by the borrower, or that is rented, will be risk weighted at 35%. In applying the 35% weight, the supervisory authorities should satisfy themselves, according to their national arrangements for the provision of housing finance, that this concessionary weight is applied restrictively for residential purposes and in accordance with strict prudential criteria, such as the existence of substantial margin of additional security over the amount of the loan based on strict valuation rules. Supervisors should increase the standard risk weight where they judge the criteria are not met.
73. National supervisory authorities should evaluate whether the risk weights in paragraph 72 are considered to be too low based on the default experience for these types of exposures in their jurisdictions. Supervisors, therefore, may require banks to increase these risk weights as appropriate.

9. Claims secured by commercial real estate

74. In view of the experience in numerous countries that commercial property lending has been a recurring cause of troubled assets in the banking industry over the past few decades, the Committee holds to the view that mortgages on commercial real estate do not, in principle, justify other than a 100% weighting of the loans secured.

10. Past due loans

75. The unsecured portion of any loan (other than a qualifying residential mortgage loan) that is past due for more than 90 days, net of specific provisions (including partial writeoffs), will be risk-weighted as follows:
- 150% risk weight when specific provisions are less than 20% of the outstanding amount of the loan;
 - 100% risk weight when specific provisions are no less than 20% of the outstanding amount of the loan;
 - 100% risk weight when specific provisions are no less than 50% of the outstanding amount of the loan, but with supervisory discretion to reduce the risk weight to 50%.
77. In addition to the circumstances described in paragraph 75, where a past due loan is fully secured by those forms of collateral that are not recognised in paragraphs 145 and 146, a 100% risk weight may apply when provisions reach 15% of the outstanding amount of the loan. These forms of collateral are not recognised elsewhere in the standardised approach. Supervisors should set strict operational criteria to ensure the quality of collateral.
78. In the case of qualifying residential mortgage loans, when such loans are past due for more than 90 days they will be risk weighted at 100%, net of specific provisions. If such loans are past due but specific provisions are no less than 20% of their outstanding amount, the risk weight applicable to the remainder of the loan can be reduced to 50% at national discretion.

13. Off-balance sheet items

82. Off-balance-sheet items under the standardised approach will be converted into credit exposure equivalents through the use of credit conversion factors (CCF). Counterparty risk weightings for OTC derivative transactions will not be subject to any specific ceiling.
83. Commitments with an original maturity up to one year and commitments with an original maturity over one year will receive a CCF of 20% and 50%, respectively. However, any commitments that are

unconditionally cancellable at any time by the bank without prior notice, or that effectively provide for automatic cancellation due to deterioration in a borrower's creditworthiness, will receive a 0% CCF.

- (i). Direct credit substitutes, e.g. general guarantees of indebtedness (including standby letters of credit serving as financial guarantees for loans and securities) and acceptances (including endorsements with the character of acceptances) will receive a CCF of 100%.
- (ii). Sale and repurchase agreements and asset sales with recourse,³⁴ where the credit risk remains with the bank will receive a CCF of 100%.

D. Standardised approach — credit risk mitigation

109. Banks use a number of techniques to mitigate the credit risks to which they are exposed. For example, exposures may be collateralised by first priority claims, in whole or in part with cash or securities, a loan exposure may be guaranteed by a third party, or a bank may buy a credit derivative to offset various forms of credit risk. Additionally banks may agree to net loans owed to them against deposits from the same counterparty.
110. Where these techniques meet the requirements for legal certainty as described in paragraph 117 and 118 below, the revised approach to CRM allows a wider range of credit risk mitigants to be recognised for regulatory capital purposes than is permitted under the 1988 Accord.

Collateralised transactions

119. A collateralised transaction is one in which:
- banks have a credit exposure or potential credit exposure; and
 - that credit exposure or potential credit exposure is hedged in whole or in part by collateral posted by a counterparty or by a third party on behalf of the counterparty.
120. Where banks take eligible financial collateral (e.g. cash or securities, more specifically defined in paragraphs 145 and 146 below), they are allowed to reduce their credit exposure to a counterparty when calculating their capital requirements to take account of the risk mitigating effect of the collateral.

Overall framework and minimum conditions

121. Banks may opt for either the simple approach, which, similar to the 1988 Accord, substitutes the risk weighting of the collateral for the risk weighting of the counterparty for the collateralised portion of the exposure (generally subject to a 20% floor), or for the comprehensive approach, which allows fuller offset of collateral against exposures, by effectively reducing the exposure amount by the value ascribed to the collateral. Banks may operate under either, but not both, approaches in the banking book, but only under the comprehensive approach in the trading book. Partial collateralisation is recognised in both approaches. Mismatches in the maturity of the underlying exposure and the collateral will only be allowed under the comprehensive approach.
122. In order for collateral to provide protection, the credit quality of the counterparty and the value of the collateral must not have a material positive correlation. For example, securities issued by the counterparty — or by any related group entity — would provide little protection and so would be ineligible.

123. Banks must have clear and robust procedures for the timely liquidation of collateral to ensure that any legal conditions required for declaring the default of the counterparty and liquidating the collateral are observed, and that collateral can be liquidated promptly.
124. Where the collateral is held by a custodian, banks must take reasonable steps to ensure that the custodian segregates the collateral from its own assets.

Simple Approach

129. In the simple approach the risk weighting of the collateral instrument collateralising or partially collateralising the exposure is substituted for the risk weighting of the counterparty. Details of this framework are provided in paragraphs 182 to 185.
130. For collateral to be recognised in the simple approach, the collateral must be pledged for at least the life of the exposure and it must be marked to market and revalued with a minimum frequency of six months. Those portions of claims collateralised by the market value of recognised collateral receive the risk weight applicable to the collateral instrument. The risk weight on the collateralised portion will be subject to a floor of 20% except under the conditions specified in paragraphs 183 to 185. The remainder of the claim should be assigned to the risk weight appropriate to the counterparty.

Suppose that a R800 million exposure to a particular counterparty is secured by collateral worth R700 million. The collateral consists of bonds issued by a BBB-rated company. The counterparty has a rating of B-. The risk weight for the counterparty is 150% and the risk weight for the collateral is 100%. The RWA applied to the exposure using the simple approach are: $(100\% \times 700) + (150\% \times \max(800-700,0)) = 850$, or R850 million. If this was not collateralised or if the collateral was not of better rating then the RWA would have been $150\% \times 800 = 1200$.

Comprehensive Approach

130. In the comprehensive approach, when taking collateral, banks will need to calculate their adjusted exposure to a counterparty for capital adequacy purposes in order to take account of the effects of that collateral. Using haircuts, banks are required to adjust both the amount of the exposure to the counterparty and the value of any collateral received in support of that counterparty to take account of possible future fluctuations in the value of either, occasioned by market movements. This will produce volatility adjusted amounts for both exposure and collateral. Unless either side of the transaction is cash, the volatility adjusted amount for the exposure will be higher than the exposure and for the collateral it will be lower.
131. Additionally where the exposure and collateral are held in different currencies an additional downwards adjustment must be made to the volatility adjusted collateral amount to take account of possible future fluctuations in exchange rates.
132. For a collateralised transaction, the exposure amount after risk mitigation is calculated as follows: $\$E^* = E - C\$$ where $\$E^*$ is the exposure value after risk mitigation, $\$E$ is the exposure value after risk mitigation and $\$C$ is the the exposure value after risk mitigation.
133. The exposure amount after risk mitigation will be multiplied by the risk weight of the counterparty to obtain the risk-weighted asset amount for the collateralised transaction.

Consider the comprehensive approach next. Assume that the adjustment to exposure to allow for possible future increases in the exposure is +10% and the adjustment to the collateral to allow for possible future decreases in its value is -15%. The new exposure is: $(1,1 \times 800) - (0,85 \times 700) = 285$, or R285 million and a risk weight of 150% is applied to this exposure to give risk-adjusted assets equal to R427,5 million.

III. Credit Risk – The Internal Ratings-Based Approach

A. Overview

211. This section of the Framework describes the IRB approach to credit risk. Subject to certain minimum conditions and disclosure requirements, banks that have received supervisory approval to use the IRB approach may rely on their own internal estimates of risk components in determining the capital requirement for a given exposure. The risk components include measures of the probability of default (PD), loss given default (LGD), the exposure at default (EAD), and effective maturity (M). In some cases, banks may be required to use a supervisory value as opposed to an internal estimate for one or more of the risk components.

212. The IRB approach is based on measures of unexpected losses (UL) and expected losses (EL). The risk-weight functions produce capital requirements for the UL portion. Expected losses are treated separately, as outlined in paragraph 43 and Section III.G.

B. Mechanics of the IRB approach

1. Categorisation of exposures

215. Under the IRB approach, banks must categorise banking-book exposures into broad classes of assets with different underlying risk characteristics, subject to the definitions set out below. The classes of assets are (a) corporate, (b) sovereign, (c) bank, (d) retail, and (e) equity. Within the corporate asset class, five sub-classes of specialised lending are separately identified. Within the retail asset class, three sub-classes are separately identified. Within the corporate and retail asset classes, a distinct treatment for purchased receivables may also apply provided certain conditions are met.

216. The classification of exposures in this way is broadly consistent with established bank practice. However, some banks may use different definitions in their internal risk management and measurement systems.

Corporate Exposure Definition

1. In general, a corporate exposure is defined as a debt obligation of a corporation, partnership, or proprietorship. Banks are permitted to distinguish separately exposures to small- and medium-sized entities (SME), as defined in paragraph 273.
2. Such lending possesses all the following characteristics, either in legal form or economic substance:
 - The exposure is typically to an entity (often a special purpose entity (SPE)) which was created specifically to finance and/or operate physical assets;
 - The borrowing entity has little or no other material assets or activities, and therefore little or no independent capacity to repay the obligation, apart from the income that it receives from the asset(s) being financed;

- The terms of the obligation give the lender a substantial degree of control over the asset(s) and the income that it generates; and
 - As a result of the preceding factors, the primary source of repayment of the obligation is the income generated by the asset(s), rather than the independent capacity of a broader commercial enterprise.
220. The five sub-classes of specialised lending are project finance, object finance, commodities finance, income-producing real estate, and high-volatility commercial real estate.
221. Project finance (PF) is a method of funding in which the lender looks primarily to the revenues generated by a single project, both as the source of repayment and as security for the exposure. This type of financing is usually for large, complex and expensive installations that might include, for example, power plants, chemical processing plants, mines, transportation infrastructure, environment, and telecommunications infrastructure. Project finance may take the form of financing of the construction of a new capital installation, or refinancing of an existing installation, with or without improvements.
222. In such transactions, the lender is usually paid solely or almost exclusively out of the money generated by the contracts for the facility's output, such as the electricity sold by a power plant. The borrower is usually an SPE that is not permitted to perform any function other than developing, owning, and operating the installation. The consequence is that repayment depends primarily on the project's cash flow and on the collateral value of the project's assets. In contrast, if repayment of the exposure depends primarily on a well established, diversified, credit-worthy, contractually obligated end user for repayment, it is considered a secured exposure to that end-user.
223. Object finance (OF) refers to a method of funding the acquisition of physical assets (e.g. ships, aircraft, satellites, railcars, and fleets) where the repayment of the exposure is dependent on the cash flows generated by the specific assets that have been financed and pledged or assigned to the lender. A primary source of these cash flows might be rental or lease contracts with one or several third parties. In contrast, if the exposure is to a borrower whose financial condition and debt-servicing capacity enables it to repay the debt without undue reliance on the specifically pledged assets, the exposure should be treated as a collateralised corporate exposure.
224. Commodities finance (CF) refers to structured short-term lending to finance reserves, inventories, or receivables of exchange-traded commodities (e.g. crude oil, metals, or crops), where the exposure will be repaid from the proceeds of the sale of the commodity and the borrower has no independent capacity to repay the exposure. This is the case when the borrower has no other activities and no other material assets on its balance sheet. The structured nature of the financing is designed to compensate for the weak credit quality of the borrower. The exposure's rating reflects its self-liquidating nature and the lender's skill in structuring the transaction rather than the credit quality of the borrower.
225. Income-producing real estate (IPRE) refers to a method of providing funding to real estate (such as, office buildings to let, retail space, multifamily residential buildings, industrial or warehouse space, and hotels) where the prospects for repayment and recovery on the exposure depend primarily on the cash flows generated by the asset. The primary source of these cash flows would generally be lease or rental payments or the sale of the asset. The borrower may be, but is not required to be, an SPE, an operating company focused on real estate construction or holdings, or an operating company with sources of revenue other than real estate. The distinguishing characteristic of IPRE versus other corporate exposures that are collateralised by real estate is the strong positive correlation between the prospects for repayment of the exposure and the prospects for recovery in the event of default, with both depending primarily on the cash flows generated by a property.
226. High-volatility commercial real estate (HVCRE) lending is the financing of commercial real estate that exhibits higher loss rate volatility (i.e. higher asset correlation)

Sovereign Exposure Definition

229. This asset class covers all exposures to counterparties treated as sovereigns under the standardised approach. This includes sovereigns (and their central banks), certain PSEs identified as sovereigns in the standardised approach, MDBs that meet the criteria for a 0% risk weight under the standardised approach, and the entities referred to in paragraph 56.

Bank Exposure Definition

230. This asset class covers exposures to banks and those securities firms outlined in paragraph 65. Bank exposures also include claims on domestic PSEs that are treated like claims on banks under the standardised approach, and MDBs that do not meet the criteria for a 0% risk weight under the standardised approach.

Retail Exposure Definition

231. An exposure is categorised as a retail exposure if it meets all of the following criteria: Nature of borrower or low value of individual exposures

- Exposures to individuals — such as revolving credits and lines of credit (e.g. credit cards, overdrafts, and retail facilities secured by financial instruments) as well as personal term loans and leases (e.g. instalment loans, auto loans and leases, student and educational loans, personal finance, and other exposures with similar characteristics) — are generally eligible for retail treatment regardless of exposure size, although supervisors may wish to establish exposure thresholds to distinguish between retail and corporate exposures.
- Residential mortgage loans (including first and subsequent liens, term loans and revolving home equity lines of credit) are eligible for retail treatment regardless of exposure size so long as the credit is extended to an individual that is an owneroccupier of the property (with the understanding that supervisors exercise reasonable flexibility regarding buildings containing only a few rental units — otherwise they are treated as corporate). Loans secured by a single or small number of condominium or co-operative residential housing units in a single building or complex also fall within the scope of the residential mortgage category. National supervisors may set limits on the maximum number of housing units per exposure.
- Loans extended to small businesses and managed as retail exposures are eligible for retail treatment provided the total exposure of the banking group to a small business borrower (on a consolidated basis where applicable) is less than €1 million. Small business loans extended through or guaranteed by an individual are subject to the same exposure threshold.

232. An exposure is categorised as a retail exposure if it meets all of the following criteria: Large number of exposures

- The exposure must be one of a large pool of exposures, which are managed by the bank on a pooled basis. Supervisors may choose to set a minimum number of exposures within a pool for exposures in that pool to be treated as retail.
- Small business exposures below €1 million may be treated as retail exposures if the bank treats such exposures in its internal risk management systems consistently over time and in the same manner as other retail exposures. This requires that such an exposure be originated in a similar manner to other

retail exposures. Furthermore, it must not be managed individually in a way comparable to corporate exposures, but rather as part of a portfolio segment or pool of exposures with similar risk characteristics for purposes of risk assessment and quantification. However, this does not preclude retail exposures from being treated individually at some stages of the risk management process. The fact that an exposure is rated individually does not by itself deny the eligibility as a retail exposure.

233. Within the retail asset class category, banks are required to identify separately three sub-classes of exposures: (a) exposures secured by residential properties as defined above, (b) qualifying revolving retail exposures, as defined in the following paragraph, and (c) all other retail exposures.
234. All of the following criteria must be satisfied for a sub-portfolio to be treated as a qualifying revolving retail exposure (QRRE). These criteria must be applied at a sub-portfolio level consistent with the bank's segmentation of its retail activities generally. Segmentation at the national or country level (or below) should be the general rule.
- (a) The exposures are revolving, unsecured, and uncommitted (both contractually and in practice). In this context, revolving exposures are defined as those where customers' outstanding balances are permitted to fluctuate based on their decisions to borrow and repay, up to a limit established by the bank.
 - (b) The exposures are to individuals.
 - (c) The maximum exposure to a single individual in the sub-portfolio is €100,000 or less.

2. Foundation and advanced approaches

244. For each of the asset classes covered under the IRB framework, there are three key elements:
- Risk components — estimates of risk parameters provided by banks some of which are supervisory estimates.
 - Risk-weight functions — the means by which risk components are transformed into risk-weighted assets and therefore capital requirements.
 - Minimum requirements — the minimum standards that must be met in order for a bank to use the IRB approach for a given asset class.
245. For many of the asset classes, the Committee has made available two broad approaches: a foundation and an advanced. Under the foundation approach, as a general rule, banks provide their own estimates of PD and rely on supervisory estimates for other risk components. Under the advanced approach, banks provide more of their own estimates of PD, LGD and EAD, and their own calculation of M, subject to meeting minimum standards. For both the foundation and advanced approaches, banks must always use the risk-weight functions provided in this Framework for the purpose of deriving capital requirements. The full suite of approaches is described below.
246. Under the foundation approach, banks must provide their own estimates of PD associated with each of their borrower grades, but must use supervisory estimates for the other relevant risk components. The other risk components are LGD, EAD and M.⁶⁶
247. Under the advanced approach, banks must calculate the effective maturity M and provide their own estimates of PD, LGD and EAD.

248. For retail exposures, banks must provide their own estimates of PD, LGD and EAD. There is no distinction between a foundation and advanced approach for this asset class.

3. Adoption of the IRB approach across asset classes

256. Once a bank adopts an IRB approach for part of its holdings, it is expected to extend it across the entire banking group. The Committee recognises however, that, for many banks, it may not be practicable for various reasons to implement the IRB approach across all material asset classes and business units at the same time. Furthermore, once on IRB, data limitations may mean that banks can meet the standards for the use of own estimates of LGD and EAD for some but not all of their asset classes/business units at the same time.
257. As such, supervisors may allow banks to adopt a phased rollout of the IRB approach across the banking group. The phased rollout includes (i) adoption of IRB across asset classes within the same business unit (or in the case of retail exposures across individual sub-classes); (ii) adoption of IRB across business units in the same banking group; and (iii) move from the foundation approach to the advanced approach for certain risk components. However, when a bank adopts an IRB approach for an asset class within a particular business unit (or in the case of retail exposures for an individual sub-class), it must apply the IRB approach to all exposures within that asset class (or sub-class) in that unit.
258. A bank must produce an implementation plan, specifying to what extent and when it intends to roll out IRB approaches across significant asset classes (or sub-classes in the case of retail) and business units over time. The plan should be exacting, yet realistic, and must be agreed with the supervisor. It should be driven by the practicality and feasibility of moving to the more advanced approaches, and not motivated by a desire to adopt a Pillar 1 approach that minimises its capital charge. During the roll-out period, supervisors will ensure that no capital relief is granted for intra-group transactions which are designed to reduce a banking group's aggregate capital charge by transferring credit risk among entities on the standardised approach, foundation and advanced IRB approaches. This includes, but is not limited to, asset sales or cross guarantees.
259. Some exposures in non-significant business units as well as asset classes (or subclasses in the case of retail) that are immaterial in terms of size and perceived risk profile may be exempt from the requirements in the previous two paragraphs, subject to supervisory approval. Capital requirements for such operations will be determined according to the standardised approach, with the national supervisor determining whether a bank should hold more capital under Pillar 2 for such positions.
260. Banks adopting an IRB approach are expected to continue to employ an IRB approach. A voluntary return to the standardised or foundation approach is permitted only in extraordinary circumstances, such as divestiture of a large fraction of the bank's credit related business, and must be approved by the supervisor.

C. UL for corporate, sovereign, and bank exposures

270. Section III.C presents the method of calculating the unexpected loss (UL) capital requirements for corporate, sovereign and bank exposures.

1. RWAs

271. The derivation of risk-weighted assets is dependent on estimates of the PD, LGD, EAD and, in some cases, effective maturity (M), for a given exposure.

272. Throughout this section, PD and LGD are measured as decimals, and EAD is measured as currency (e.g. euros), except where explicitly noted otherwise. For exposures not in default, the formula for calculating risk-weighted assets is

$$\text{Correlation } \rho = R = 0.12 \sqrt{\frac{1 - e^{-50PD}}{1 - e^{-50}}} + 0.24(1 - \sqrt{\frac{1 - e^{-50PD}}{1 - e^{-50}}})$$

$$\text{Maturity Adjustment } b = [0.11852 - 0.05478 \ln(\text{PD})]^2$$

$$K = [N(\sqrt{\frac{G(PD) + \sqrt{R}}{1 - R}} \times G(0.999)) - PD] \times \text{LGD} - PD \times \text{LGD}$$

$$\text{RWA} = 12.5 \times K \times \text{EAD}$$

The capital requirement (K) for a defaulted exposure is equal to the greater of zero and the difference between its LGD (described in paragraph 468) and the bank's best estimate of expected loss (described in paragraph 471). The risk-weighted asset amount for the defaulted exposure is the product of K, 12.5, and the EAD.

D. UL for retail exposures

326. Section D presents in detail the method of calculating the UL capital requirements for retail exposures.

1. RWAs

327. There are three separate risk-weight functions for retail exposures, as defined in paragraphs 328 to 330. Risk weights for retail exposures are based on separate assessments of PD and LGD as inputs to the risk-weight functions. None of the three retail risk-weight functions contains an explicit maturity adjustment. Throughout this section, PD and LGD are measured as decimals, and EAD is measured as currency (e.g. euros).

328. For exposures defined in paragraph 231 that are not in default and are secured or partly secured by residential mortgages, risk weights will be assigned based on the following formula:

$$\text{Correlation } \rho = R = 0.15$$

$$K = [\text{LGD} \times N(\sqrt{\frac{G(PD) + \sqrt{R}}{1 - R}} \times G(0.999)) - PD] \times \text{EAD}$$

$$\text{RWA} = 12.5 \times K \times \text{EAD}$$

The capital requirement (K) for a defaulted exposure is equal to the greater of zero and the difference between its LGD (described in paragraph 468) and the bank's best estimate of expected loss (described in paragraph 471). The risk-weighted asset amount for the defaulted exposure is the product of K, 12.5, and the EAD.

329. For qualifying revolving retail exposures as defined in paragraph 234 that are not in default, risk weights are defined based on the following formula:

$$\text{Correlation } \rho = R = 0$$

$$K = [\text{LGD} \times N(\sqrt{\frac{G(PD) + \sqrt{R}}{1 - R}} \times G(0.999)) - PD] \times \text{EAD}$$

$$\text{RWA} = 12.5 \times K \times \text{EAD}$$

The capital requirement (K) for a defaulted exposure is equal to the greater of zero and the difference between its LGD (described in paragraph 468) and the bank's best estimate of expected loss (described in paragraph 471). The risk-weighted asset amount for the defaulted exposure is the product of K, 12.5, and the EAD.

G. Treatment of Expected Losses and Recognition of Provisions

374. Section III.G discusses the method by which the difference between provisions (e.g. specific provisions, portfolio-specific general provisions such as country risk provisions or general provisions) and expected losses may be included in or must be deducted from regulatory capital, as outlined in paragraph 43.

1. Calculation of expected losses

375. A bank must sum the EL amount (defined as EL multiplied by EAD) associated with its exposures (excluding the EL amount associated with equity exposures under the PD/LGD approach and securitisation exposures) to obtain a total EL amount.

2. Calculation of provisions

380. Total eligible provisions are defined as the sum of all provisions (e.g. specific provisions, partial write-offs, portfolio-specific general provisions such as country risk provisions or general provisions) that are attributed to exposures treated under the IRB approach. In addition, total eligible provisions may include any discounts on defaulted assets. Specific provisions set aside against equity and securitisation exposures must not be included in total eligible provisions.

3. Treatment of EL and provisions

384. As specified in paragraph 43, banks using the IRB approach must compare the total amount of total eligible provisions (as defined in paragraph 380) with the total EL amount as calculated within the IRB approach (as defined in paragraph 375). In addition, paragraph 42 outlines the treatment for that portion of a bank that is subject to the standardised approach to credit risk when the bank uses both the standardised and IRB approaches.

385. Where the calculated EL amount is lower than the provisions of the bank, its supervisors must consider whether the EL fully reflects the conditions in the market in which it operates before allowing the difference to be included in Tier 2 capital. If specific provisions exceed the EL amount on defaulted assets this assessment also needs to be made before using the difference to offset the EL amount on non-defaulted assets.

Part 3: The Second Pillar – Supervisory Review Process

719. This section discusses the key principles of supervisory review, risk management guidance and supervisory transparency and accountability produced by the Committee with respect to banking risks, including guidance relating to, among other things, the treatment of interest rate risk in the banking book, credit risk (stress testing, definition of default, residual risk, and credit concentration risk), operational risk, enhanced cross-border communication and cooperation, and securitisation.

I. Importance of supervisory review

720. The supervisory review process of the Framework is intended not only to ensure that banks have adequate capital to support all the risks in their business, but also to encourage banks to develop and use better risk management techniques in monitoring and managing their risks.
721. The supervisory review process recognises the responsibility of bank management in developing an internal capital assessment process and setting capital targets that are commensurate with the bank's risk profile and control environment. In the Framework, bank management continues to bear responsibility for ensuring that the bank has adequate capital to support its risks beyond the core minimum requirements.
722. Supervisors are expected to evaluate how well banks are assessing their capital needs relative to their risks and to intervene, where appropriate. This interaction is intended to foster an active dialogue between banks and supervisors such that when deficiencies are identified, prompt and decisive action can be taken to reduce risk or restore capital. Accordingly, supervisors may wish to adopt an approach to focus more intensely on those banks with risk profiles or operational experience that warrants such attention.
723. The Committee recognises the relationship that exists between the amount of capital held by the bank against its risks and the strength and effectiveness of the bank's risk management and internal control processes. However, increased capital should not be viewed as the only option for addressing increased risks confronting the bank. Other means for addressing risk, such as strengthening risk management, applying internal limits, strengthening the level of provisions and reserves, and improving internal controls, must also be considered. Furthermore, capital should not be regarded as a substitute for addressing fundamentally inadequate control or risk management processes.
724. There are three main areas that might be particularly suited to treatment under Pillar 2: risks considered under Pillar 1 that are not fully captured by the Pillar 1 process (e.g. credit concentration risk); those factors not taken into account by the Pillar 1 process (e.g. interest rate risk in the banking book, business and strategic risk); and factors external to the bank (e.g. business cycle effects). A further important aspect of Pillar 2 is the assessment of compliance with the minimum standards and disclosure requirements of the more advanced methods in Pillar 1, in particular the IRB framework for credit risk and the Advanced Measurement Approaches for operational risk. Supervisors must ensure that these requirements are being met, both as qualifying criteria and on a continuing basis.

II. Four key principles of supervisory review

725. The Committee has identified four key principles of supervisory review, which complement those outlined in the extensive supervisory guidance that has been developed by the Committee.

Principle 1

Banks should have a process for assessing their overall capital adequacy in relation to their risk profile and a strategy for maintaining their capital levels.

727. The five main features of a rigorous process are as follows:

- Board and senior management oversight;
- Sound capital assessment;
- Comprehensive assessment of risks;
 - Credit risk
 - Op risk
 - Market risk

- Interest rate risk in the banking book
- Liquidity risk
- Other risks
- Monitoring and reporting; and
- Internal control review.

Principle 2

Supervisors should review and evaluate banks' internal capital adequacy assessments and strategies, as well as their ability to monitor and ensure their compliance with regulatory capital ratios. Supervisors should take appropriate supervisory action if they are not satisfied with the result of this process.

Principle 3

Supervisors should expect banks to operate above the minimum regulatory capital ratios and should have the ability to require banks to hold capital in excess of the minimum.

Principle 4

Supervisors should seek to intervene at an early stage to prevent capital from falling below the minimum levels required to support the risk characteristics of a particular bank and should require rapid remedial action if capital is not maintained or restored.

III. Specific issues to be addressed under the supervisory review process

761. The Committee has identified a number of important issues that banks and supervisors should particularly focus on when carrying out the supervisory review process. These issues include some key risks which are not directly addressed under Pillar 1 and important assessments that supervisors should make to ensure the proper functioning of certain aspects of Pillar 1.

A. Interest rate risk in the banking book

762. The Committee remains convinced that interest rate risk in the banking book is a potentially significant risk which merits support from capital. However, comments received from the industry and additional work conducted by the Committee have made it clear that there is considerable heterogeneity across internationally active banks in terms of the nature of the underlying risk and the processes for monitoring and managing it. In light of this, the Committee has concluded that it is at this time most appropriate to treat interest rate risk in the banking book under Pillar 2 of the Framework. Nevertheless, supervisors who consider that there is sufficient homogeneity within their banking populations regarding the nature and methods for monitoring and measuring this risk could establish a mandatory minimum capital requirement.

B. Credit risk

1. Stress tests under the IRB approaches

765. A bank should ensure that it has sufficient capital to meet the Pillar 1 requirements and the results (where a deficiency has been indicated) of the credit risk stress test performed as part of the Pillar 1 IRB

minimum requirements (paragraphs 434 to 437). Supervisors may wish to review how the stress test has been carried out. The results of the stress test will thus contribute directly to the expectation that a bank will operate above the Pillar 1 minimum regulatory capital ratios. Supervisors will consider whether a bank has sufficient capital for these purposes. To the extent that there is a shortfall, the supervisor will react appropriately. This will usually involve requiring the bank to reduce its risks and/or to hold additional capital/provisions, so that existing capital resources could cover the Pillar 1 requirements plus the result of a recalculated stress test.

766. Definition of default

767. Residual risk

768. Credit concentration risk

769. Counterparty credit risk

C. Operational risk

Gross income, used in the Basic Indicator and Standardised Approaches for operational risk, is only a proxy for the scale of operational risk exposure of a bank and can in some cases (e.g. for banks with low margins or profitability) underestimate the need for capital for operational risk. With reference to the Committee document on Sound Practices for the Management and Supervision of Operational Risk (February 2003), the supervisor should consider whether the capital requirement generated by the Pillar 1 calculation gives a consistent picture of the individual bank's operational risk exposure, for example in comparison with other banks of similar size and with similar operations.

D. Market risk

3. Stress testing under the internal models approach

778(iii). A bank must ensure that it has sufficient capital to meet the minimum capital requirements set out in paragraphs 718(Lxx) to 718(xciv) and to cover the results of its stress testing required by paragraph 718(Lxxiv) (g), taking into account the principles set forth in paragraphs 738(ii) and 738(iv). Supervisors will consider whether a bank has sufficient capital for these purposes, taking into account the nature and scale of the bank's trading activities and any other relevant factors such as valuation adjustments made by the bank. To the extent that there is a shortfall, or if supervisors are not satisfied with the premise upon which the bank's assessment of internal market risk capital adequacy is based, supervisors will take the appropriate measures. This will usually involve requiring the bank to reduce its risk exposures and/or to hold an additional amount of capital, so that its overall capital resources at least cover the Pillar 1 requirements plus the result of a stress test acceptable to the supervisor.

Part 4: The Third Pillar – Market Discipline

I. General considerations

B. Guiding principles

809. The purpose of Pillar 3 — market discipline is to complement the minimum capital requirements (Pillar 1) and the supervisory review process (Pillar 2). The Committee aims to encourage market discipline by

developing a set of disclosure requirements which will allow market participants to assess key pieces of information on the scope of application, capital, risk exposures, risk assessment processes, and hence the capital adequacy of the institution. The Committee believes that such disclosures have particular relevance under the Framework, where reliance on internal methodologies gives banks more discretion in assessing capital requirements.

810. In principle, banks' disclosures should be consistent with how senior management and the board of directors assess and manage the risks of the bank. Under Pillar 1, banks use specified approaches/methodologies for measuring the various risks they face and the resulting capital requirements. The Committee believes that providing disclosures that are based on this common framework is an effective means of informing the market about a bank's exposure to those risks and provides a consistent and understandable disclosure framework that enhances comparability.

D. Interaction with accounting disclosures

813. The Committee recognises the need for a Pillar 3 disclosure framework that does not conflict with requirements under accounting standards, which are broader in scope. The Committee has made a considerable effort to see that the narrower focus of Pillar 3, which is aimed at disclosure of bank capital adequacy, does not conflict with the broader accounting requirements. Going forward, the Committee intends to maintain an ongoing relationship with the accounting authorities, given that their continuing work may have implications for the disclosures required in Pillar 3. The Committee will consider future modifications to Pillar 3 as necessary in light of its ongoing monitoring of this area and industry developments.
814. Management should use its discretion in determining the appropriate medium and location of the disclosure. In situations where the disclosures are made under accounting requirements or are made to satisfy listing requirements promulgated by securities regulators, banks may rely on them to fulfil the applicable Pillar 3 expectations. In these situations, banks should explain material differences between the accounting or other disclosure and the supervisory basis of disclosure. This explanation does not have to take the form of a line by line reconciliation.
815. For those disclosures that are not mandatory under accounting or other requirements, management may choose to provide the Pillar 3 information through other means (such as on a publicly accessible internet website or in public regulatory reports filed with bank supervisors), consistent with requirements of national supervisory authorities.

E. Materiality

817. A bank should decide which disclosures are relevant for it based on the materiality concept. Information would be regarded as material if its omission or misstatement could change or influence the assessment or decision of a user relying on that information for the purpose of making economic decisions. This definition is consistent with International Accounting Standards and with many national accounting frameworks.

Frequency

818. The disclosures set out in Pillar 3 should be made on a semi-annual basis, subject to the following exceptions. Qualitative disclosures that provide a general summary of a bank's risk management objectives and policies, reporting system and definitions may be published on an annual basis. In recognition of the increased risk sensitivity of the Framework and the general trend towards more

frequent reporting in capital markets, large internationally active banks and other significant banks (and their significant bank subsidiaries) must disclose their Tier 1 and total capital adequacy ratios, and their components,¹⁷⁴ on a quarterly basis. Furthermore, if information on risk exposure or other items is prone to rapid change, then banks should also disclose information on a quarterly basis.

II. The disclosure requirements

820. The following sections set out in tabular form the disclosure requirements under Pillar 3. Additional definitions and explanations are provided in a series of footnotes.